Software Requirements Specification

Unified Social Media Marketing & Ad Management Dashboard

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) documents the functional and non-functional requirements for a unified social media marketing dashboard. The platform is designed to consolidate organic publishing, community engagement, and paid advertisement management across multiple social networks into a single interface.

1.2 Scope

The system will function as a centralized command center for digital marketers. Core capabilities include cross-platform content scheduling, unified inbox management, API-driven ad campaign execution, and advanced analytics. The platform will differentiate itself through AI-assisted content optimization, dynamic budget reallocation for ads, and tools for managing private communities (e.g., WhatsApp, Discord), moving beyond traditional public feed management.

1.3 Target Audience

The primary users of this system are digital marketing agencies, solo entrepreneurs, and small-to-medium e-commerce brands who require an integrated view of their organic and paid social media efforts without the overhead of enterprise-level software suites.

2. Overall Description

2.1 Product Perspective

The dashboard is a web-based SaaS (Software as a Service) application. It acts as an intermediary layer between the user and various external social media APIs (Meta Graph API, TikTok API, X API, LinkedIn

Marketing Developer Platform). The system relies heavily on OAuth for authentication and real-time data synchronization with these external services.

2.2 System Features

2.2.1 Unified Inbox & Engagement

- **Requirement:** The system shall aggregate Direct Messages, comments, and mentions from connected Facebook, Instagram, TikTok, LinkedIn, and X accounts into a single chronological feed.
- **Requirement:** Users shall be able to reply to messages directly from the unified inbox, with the system routing the response to the appropriate native platform API.
- **Requirement:** The system shall support tagging and filtering of inbox items by sentiment, platform, or custom user-defined labels.

2.2.2 Publishing & Scheduling

- **Requirement:** The system shall provide a visual calendar interface for planning and scheduling content.
- **Requirement:** The system shall support multi-platform posting, allowing a single piece of content to be customized (e.g., aspect ratio, character limits) for different networks within the same workflow.
- **Requirement:** The system shall include an "AI Content Slicer" tool capable of generating platform-specific short-form content (e.g., a Twitter thread or TikTok script) from a user-provided long-form URL.

2.2.3 Paid Ad Management

- **Requirement:** The system shall integrate with Meta Ads and TikTok Ads APIs to allow users to launch, pause, and adjust daily budgets for campaigns directly from the dashboard.
- **Requirement:** The system shall feature a "Cross-Platform Ad Reallocation" tool that automatically shifts defined budgets to the platform yielding the lowest Cost Per Acquisition (CPA) based on real-time API data.

2.2.4 Analytics & Reporting

- **Requirement:** The system shall generate consolidated reports combining organic reach metrics with paid campaign ROI.
- **Requirement:** The system shall monitor engagement velocity to power a "Creative Fatigue Tracker," alerting users when specific assets experience a statistically significant drop in scroll-stop or watch-through rates.

3. Non-Functional Requirements

> **Critical Constraint:** Due to the volatile nature of third-party platform rules, the system architecture must prioritize modularity. The failure or rate-limiting of one API (e.g., X API) must not degrade the performance of other integrations (e.g., Meta or TikTok).

Category	Requirement Specification
Performance	The unified inbox must reflect new incoming messages from platform APIs within 2 minutes of the native event.
Scalability	The backend architecture must support microservices to scale video processing and AI token generation independently of core API polling services.
Security	All API tokens and OAuth credentials must be encrypted at rest. The application must comply with the data privacy policies mandated by Meta, TikTok, and other integrated platforms.
Reliability	The core application must maintain 99.9% uptime, excluding scheduled maintenance. Background retry logic must be implemented for failed API post schedules.

4. System Interfaces

The application will interface primarily with RESTful and GraphQL APIs provided by external social networks. The frontend will be a responsive web application designed for desktop use, with a simplified mobile-responsive view for on-the-go inbox management. Cloud infrastructure (e.g., AWS or GCP) will be utilized for database hosting (PostgreSQL for relational data, Redis for caching) and object storage for media assets.